

MotionIQ: Athlete Video Upload & Screening Workflow Specification

Complete Technical Guide from Video File Upload to Computer Vision, Feature Mathematics, ML Scoring & PDF Reports

1. Executive Workflow Summary

MotionIQ provides a sensorless, non-invasive biomechanical movement screening platform. Athletes or coaches upload optical video recordings of movement exercises (such as squats, vertical jumps, or cutting drills), which are automatically converted into 3D skeletal trajectories, evaluated mathematically for kinematic risk deviations, and scored by Machine Learning classifiers to prevent injuries before they occur.

2. Detailed 5-Phase End-to-End Processing Architecture

Phase 1: Athlete Ingestion & Video Upload

The athlete selects their profile (Name, Sport, Position, Height, Weight, Injury History) and uploads a movement video file (.mp4, .mov) or starts live webcam capture. The system pre-validates video frame rate (target 60 FPS), aspect ratio, and lighting integrity.

Phase 2: Computer Vision & Skeletal Pose Tracking

OpenCV ingests the video and extracts RGB frames. Google MediaPipe Pose tracks 33 3D skeletal landmarks per frame: $P = (x, y, z, \text{visibility})$. Normalized spatial coordinates are converted into sequential landmark time-series data.

Phase 3: Mathematical Feature Derivation Engine

The engine computes mathematical joint angles using vector dot-products: $\text{acos}((\mathbf{BA} \cdot \mathbf{BC}) / (|\mathbf{BA}| * |\mathbf{BC}|))$. It derives Range of Motion (ROM), Left/Right Bilateral Limb Symmetry %, Spinal Trunk Lean Angle, and Kinematic Fatigue Index.

Phase 4: Machine Learning Inference & Multi-Class Scoring

Features are compiled into a 13-dimensional input vector X_{input} passed to trained XGBoost & Random Forest Ensembles. The model evaluates Injury Risk % (0-100), classifies Risk Level (LOW, MODERATE, HIGH, CRITICAL), and identifies specific injury categories (ACL, Ankle, Knee).

Phase 5: Output Generation, Rehabilitation & Dashboard UI

Generates a downloadable clinical PDF report, prescribes multi-phase corrective physical therapy exercises, updates the interactive 3D Skeletal Canvas, populates the Biomechanical Body Heatmap, and triggers AI Voice Briefings.

3. Summary Table: Inputs, Actions & Outputs per Processing Phase

Workflow Phase	Input Data	System Action & Algorithms	Generated Output
1. Athlete Upload	Raw Video File (.mp4) + Profile Metadata	Client-side validation & REST API POST /api/videos/upload	Uploaded video stored in temp server buffer
2. Pose Tracking	Raw RGB Video Frames	OpenCV frame extraction + MediaPipe 33 landmark tracking	Time-series keypoint coordinates: $P(x, y, z, \text{vis})$
3. Feature Derivation	Landmark time-series data	Vector dot-product angles, ROM, Symmetry %, Trunk Lean, Fatigue Index	Kinematic metric dictionary (Angles, ROM, Symmetry)
4. ML Scoring	13D Feature Vector X_{input}	Supervised XGBoost & Random Forest Inference	Injury Risk Score % + Specific Category Predictions
5. Output & Report	ML Predictions + Biomechanics	PDF generator + Exercise prescriptor + UI socket update	Downloadable Clinical PDF + 3D Skeletal Canvas + AI Voice

4. Concrete Numerical Example: Bilateral Squat Screening

Example Input: Athlete performs a 3-second bilateral squat (180 frames @ 60 FPS).

- **Frame 90 (Peak Flexion):** Left Knee Angle = 78.5 deg | Right Knee Angle = 94.2 deg.
- **Range of Motion:** Left ROM = 93.5 deg | Right ROM = 75.8 deg.
- **Calculated Symmetry Score:** $100 * (1 - |93.5 - 75.8| / 93.5) = 81.1\%$ (**18.9% Asymmetry Deficit**).
- **Trunk Lean Angle:** 28.4 deg (Exceeds 25.0 deg postural limit).
- **Fatigue Index:** 31.5 pts (Exceeds 25.0 pts consistency threshold).
- **ML XGBoost Score:** **78.4% (HIGH RISK)**.
- **Prescription Output:** Single-Leg Eccentric Squats (3x10), Gluteus Medius Band Walks (4x15), Nordic Hamstring Curls (3x8).